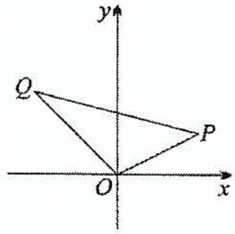
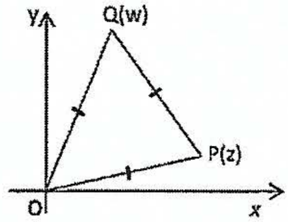
Complex Number 8

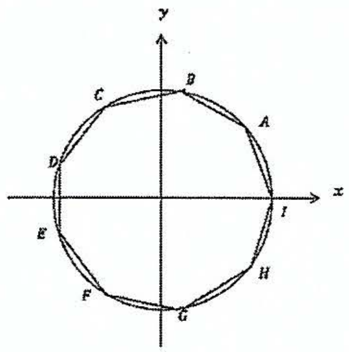
1. Suppose that the complex number z has modulus one, and that $0 < \arg z < \frac{\pi}{2}$. Prove that $2\arg(z+1) = \arg z$.
2. The vertices of the quadrilateral $ABCD$ in the complex plane represent the complex numbers z_1, z_2, z_3 and z_4 respectively.
   a. If $z_1 - z_2 = z_4 - z_3$, show that the quadrilateral $ABCD$ is a parallelogram.
   b. If $z_1 - z_2 = z_4 - z_3$ and $z_1 - z_3 = i(z_4 - z_2)$, show that $ABCD$ is a square.
3. a. Prove that for any complex number z, $|z|^2 = z\bar{z}$
   b. Hence prove that for any complex number z_1 and z_2: $|z_1+z_2|^2 + |z_1-z_2|^2 = 2(|z_1|^2+|z_2|^2)$
   c. Explain this result geometrically.
4. In the diagram on the right, the point P and Q represent the complex number z and w respectively.
   a. Explain why $|z-w| \le |z| + |w|$.
   b. Indicate on the diagram the point R representing $z+w$.
   c. What type of quadrilateral is $OPRQ$?
   d. If $|z-w| = |z+w|$, what can be said about the complex number $\frac{w}{z}$?



5. If $|z_1 - z_2| = |z_1 + z_2|$, what can be said about the complex number $\frac{z_1}{z_2}$?
6. In the Argand diagram, OPQ is an equilateral triangle. P represents the complex number z and Q represents the complex number w. Show that $w^3 + z^3 = 0$.



7. a. Solve $z^9 - 1 = 0$.
   b. If w is the root of $z^9 - 1 = 0$ with the smallest positive argument, show that $w^2, w^3, w^4, w^5, w^6, w^7$ and w^8 are non real roots of this equation.
   c. On the diagram below, $\overrightarrow{OA} = w$, $\overrightarrow{OB} = w^2$, $\overrightarrow{OC} = w^4$, $\overrightarrow{OE} = w^5$, $\overrightarrow{OF} = w^6$, $\overrightarrow{OG} = w^7$, $\overrightarrow{OH} = w^8$ and $\overrightarrow{OI} = 1$.



   i. Find the area of the regular nonagon formed by joining A, B, C, D, E, F, G, H and I. (Answer to 2 decimal places.)
   ii. Prove $\arg\left(\frac{w-w^6}{w^8-w^5}\right) = \frac{\pi}{3}$.

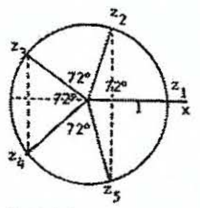
   d. i. Show that $1 + w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = 0$.
   ii. Hence or otherwise show that $(w^4+w^5)(w^2+w^7) + (w^3+w^6)(w^2+w^7) = -1$.
   iii. Hence or otherwise show that $\cos\frac{4\pi}{9}\left(2\cos\frac{8\pi}{9} - 1\right) = -\frac{1}{2}$.

8. If ω is a complex root of $z^5 - 1 = 0$, show that $\omega^2, \omega^3,$ and ω^4 are the other complex roots.
    a. Prove that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$
    b. Find the quadratic equations whose roots are $\alpha = \omega + \omega^4$ and $\beta = \omega^2 + \omega^3$ (Hint: use $\omega^5 = 1$ to reduce ω^6 and ω^7)
    c. Show the roots of $z^5 - 1 = 0$ in an Argand diagram
    d. Find the area of the pentagon formed by the roots
9. If ω is the complex root of $z^6 - 1 = 0$ with the smallest positive argument, then show that the other roots are $\omega^2, \omega^3, \omega^4, \omega^5$. Prove that:
    a. $1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 = 0$
    b. Find all the roots in the form $a + ib$ and indicate these roots in an Argand diagram. Find the area of the hexagon formed by the roots.
    c. Find the two quadratic equations whose roots are
        i. ω and ω^5
        ii. ω^2 and ω^4
    d. Using part c., show that
        i. $z^6 - 1 = (z-1)(z+1)[(z-w)(z-w^5)][(z-\omega^2)(z-\omega^4)] = (z^2-1)(z^2+z+1)(z^2-z+1)$
        ii. The roots of $z^4 + z^2 + 1 = 0$ are $\omega, \omega^2, \omega^4$ and ω^5
10. Show that if ω is one complex root of equation $z^n - 1 = 0,$ then
    a. $z^n - 1 = (z-1)(z-\omega)(z-\omega^2) \ldots (z-\omega^{n-1})$
    b. Deduce from part (a) that: $z^{n-1} + z^{n-2} + \cdots + z + 1 = (z-\omega)(z-\omega^2) \ldots (z-\omega^{n-1})$
    c. Hence, show $(1-\omega)(1-\omega^2) \ldots (1-\omega^{n-1}) = n$
11. On an Argand diagram P and Q represent the complex numbers z_1 and z_2 respectively. OPQ is an equilateral triangle in which $|z_1| = |z_2| = 1$.
    a. Write an expression in terms of z_1 and z_2 for the complex number represented by the vector PQ.
    b. Find $|z_1 + z_2|$
    c. Find a possible value of k if $\frac{z_1+z_2}{z_1-z_2} = k$
12. Given $z = r(\cos\theta + i\sin\theta)$, where $z \neq 0$:
    a. Show that $z\bar{z}$ is real.
    b. Use De Moivre's Theorem to show that $z^n + \bar{z}^n$ is real for all integers $n \geq 1$.
    c. Show that $\frac{z}{\bar{z}} + \frac{\bar{z}}{z}$ is real.
    d. Hence, show that $-2 \leq \frac{z}{\bar{z}} + \frac{\bar{z}}{z} \leq 2$.

Answer:

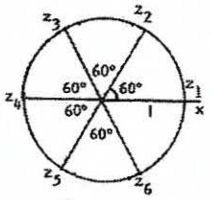
1. c. The sum of the squares of the diagonals of a parallelogram is equal to 2 times the sum of the squares of its sides.
4. c. Parallelogram d. $\arg\frac{w}{z} = \frac{\pi}{2}$, so $\frac{w}{z}$ is purely imaginary
5. $\frac{z_1}{z_2}$ is pure imaginary.
7. a. $z = \text{cis}\frac{2k\pi}{9}, k = 0, 1, 2, \ldots$ c. i. Area $= 2.89$ square units
8. b. $z^2 + z - 1 = 0$

c.



d. 2.38

9. b. $\pm 1, \pm\frac{1}{2}.\pm\frac{\sqrt{3}}{2}i$, area $= \frac{3\sqrt{3}}{2}$



c. i. $z^2 - z + 1 = 0$ ii. $z^2 + z + 1 = 0$

11. a. $PQ = z_2 - z_1$ b. $|z_1 + z_2| = \sqrt{3}$ c. $k = i\sqrt{3}$